

Qie Sima

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Hong Kong University of Science and Technology, Kowloon, Hong Kong.

EDUCATION

Hong Kong University of Science and Technology

Ph.D. Student in Computer Science; Supervised by [Prof. Yike Guo](#)

Kowloon, Hong Kong

Sep 2024 - Present

Research focus: Multi-modal Large Models Pretraining, VLA Pre-training, Data Auto-labeling & Mixture Optimization.

Tsinghua University

Master of Science in Computer Science; Supervised by [Prof. Huaping Liu](#)

Beijing, China

Sep 2020 - Jun 2023

Research focus: Robotic Manipulation, Visual Navigation, Imitation Learning.

Tsinghua University

Bachelor of Engineering in Engineering Mechanics (Tsien Excellence in Education Program)

Beijing, China

Sep 2016 - Jun 2020

Enrolled in the elite TEEP program under the School of Aerospace Engineering.

RESEARCH & WORK EXPERIENCE

Agibot

Embodied AI Algorithm Engineer (VLA Pre-training)

Shanghai / Beijing, China

Nov 2025 - Present

- **CoT-VLA Pre-training & Data Recipe:** Architected a Vision-Language-Action (VLA) pre-training framework based on Qwen3-VL-4B with Chain-of-Thought (CoT) injection. Proven that CoT speeds up loss convergence; optimized data recipes to achieve 97.5% baseline success rate on LIBERO and 95%+ with CoT reasoning.
- **Closed-Loop Data Curation & Annotation Engine:** Built an end-to-end, automated multi-source data engine integrating LLM APIs and local models. Streamlined filtering (OOD/corrupted frames), phase segmentation, and multi-modal tagging across 100k+ industrial and general trajectories for seamless pre-training pipeline.
- **Deployments & Long-Horizon Tasks:** Validated data mixtures on physical wheeled humanoid platforms, streamlining Dagger data curation and fine-grained hardware calibration for long-horizon industrial/indoor tasks.

AgileX Robotics

Algorithm Engineer Intern

Shenzhen, China

Jun 2025 - Oct 2025

- **Data Evaluation & Segmentation Pipeline:** Developed an OpenVLA and Qwen3-VL based dataset evaluation pipeline utilizing tri-view feature fusion and entropy-weighted quality scoring. Integrated SAM 2 for pixel-level segmentation and high-level semantic annotation.
- **Real-Robot Tabletop Manipulation:** Fine-tuned and deployed $\pi_{0.5}$ base models on custom-collected datasets for tabletop tasks (grasping, folding clothes, sorting/storage) on dual-arm setups.
- **VLA Post-Training Quantization Benchmark:** Led quantization studies (INT8/INT4, GPTQ, AWQ) on OpenVLA and RT-2-X models, building an inference speed, VRAM footprint, and task success rate benchmark for edge robot deployment.

Lenovo Research, Lenovo Inc.

Research Engineer (Mobile Intelligence)

Beijing, China

Aug 2023 - Aug 2024

- **Multimodal Foundation Models:** Trained and benchmarked multi-modal perception models for personal smart devices. Researched model compression and quantization strategies for LLaMA-2 architectures on resource-constrained mobile platforms and robotic agents.

AI Lab, ByteDance Inc.

Robotics Algorithm Research Intern (Mentored by [Dr. Tao Kong](#))

Beijing, China

May 2022 - Jul 2023

- **GR-1 Humanoid VLA Pre-training Data:** Constructed upstream pre-training datasets for humanoid manipulation by curating and adapting large-scale vision benchmarks (COCO, Visual Genome, Ego4D) to embodied settings.
- **Visual Representation Ablation:** Evaluated vision backbones (CLIP, MoCo-v3, MAE) for VLA models. Conducted multi-modal masking and ablation studies across vision, language, and temporal dimensions.
- **3D Scene Reconstruction for Home Robots:** Co-designed 3D online scene reconstruction for mobile manipulation service robots, validating mobile grasping, door opening, and switch toggling in real-world trials.

Georgia Institute of Technology

Visiting Student Researcher (Supervised by [Dr. Ye Zhao](#))

Atlanta, GA, USA

Aug 2019 - Feb 2020

- **Soft Contact Mechanics & Control:** Formulated single-point deformation mechanics for bio-compatible silicone under soft contacts. Designed an ADMM-based force feedback trajectory optimizer validated via real-world experiments.

SELECTED PUBLICATIONS

- **ProMix: Learning Optimal Data Mixtures for Robotic Imitation via Proxy-Reference Distillation:** Q. Sima, et al. Submitted to *ICRA 2026*. Introduces distributionally robust optimization via proxy models to dynamically curate heterogeneous datasets, dramatically reducing trial-and-error training costs and boosting VLA success rates by 7%–10%.
- **Smooth Computation without Input Delay: Robust Tube-Based Model Predictive Control for Robot Manipulator Planning:** Co-author. *IEEE International Conference on Robotics and Automation (ICRA 2024)*. Proposes a latency-compensated Tube-MPC formulation linearized for real-time robotic arm trajectory planning.
- **Exploring Visual Pre-training for Robot Manipulation: Datasets, Models and Methods (Vi-PRoM):** Co-author. *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2023)*. Pre-trains ResNet-50 backbones on temporal human video (Ego4D) for downstream PPO policy transfer.
- **MOMA-Force: Visual-Force Imitation for Real-World Mobile Manipulation:** Co-author. *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2023)*. Integrates visual and wrench force perception into end-to-end mobile imitation learning.
- **Embodied Referring Expression for Manipulation Question Answering in Interactive Environment:** Q. Sima, et al. *IEEE International Conference on Robotics and Automation (ICRA 2023)*. Incorporates referring expression comprehension and 3D scene graphs into AI2-THOR mobile manipulation QA.
- **Scene Graph for Embodied Exploration in Cluttered Scenario:** Q. Sima, et al. *ACM/IEEE International Conference on Human-Robot Interaction (HRI 2023)*. Abstracts non-structured object clutter into scene graph priors for physical manipulation QA.
- **Active Exploration for Answering Questions in Bin Scenarios:** Q. Sima, et al. *IEEE/ASME Transactions on Mechatronics (T-MECH 2022)*. Defines the Manipulation Question Answering (MQA) task paradigm and proposes combined IL and KG reasoning algorithms.
- **Simultaneous Trajectory Optimization and Force Control with Soft Contact Mechanics:** Co-author. *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2020)*. Real-time trajectory optimization and force control formulation under soft bio-material contact.

PATENTS & COMPETITION AWARDS

- **Granted US Patent:** Method and apparatus for robot control, device, robot, and medium (US Patent 12,521,890). Co-inventors: T. Yang, J. Ya, H. Wu, K. Sima, Q. Sima, T. Kong. Issued Jan 13, 2026.
- **Granted CN Patent:** A Robot Question-Answering Method for Object Search (CN113516055A), 2022.
- **RoboTHOR Challenge:** 2nd Place worldwide in Visual Semantic Navigation, CVPR 2020 Embodied AI Workshop.

ACADEMIC SERVICES & ACTIVITIES

- **Conference Reviewer:** IEEE International Conference on Robotics and Automation (ICRA), IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Conference on Robot Learning (CoRL), ACM/IEEE Human-Robot Interaction (HRI).
- **Competition Organization:** Co-organizer of AgiBot World Challenge 2026 at ICRA 2026.
- **Oral & Poster Presentations:** Presented Oral report at ICRA 2024 (Yokohama); Poster presentations at ICRA 2026 (Vienna), ICRA 2023 (London), IROS 2023 (Detroit), and IROS 2020 (Online); Oral presenter at ECSSMET 2018 (Netherlands).